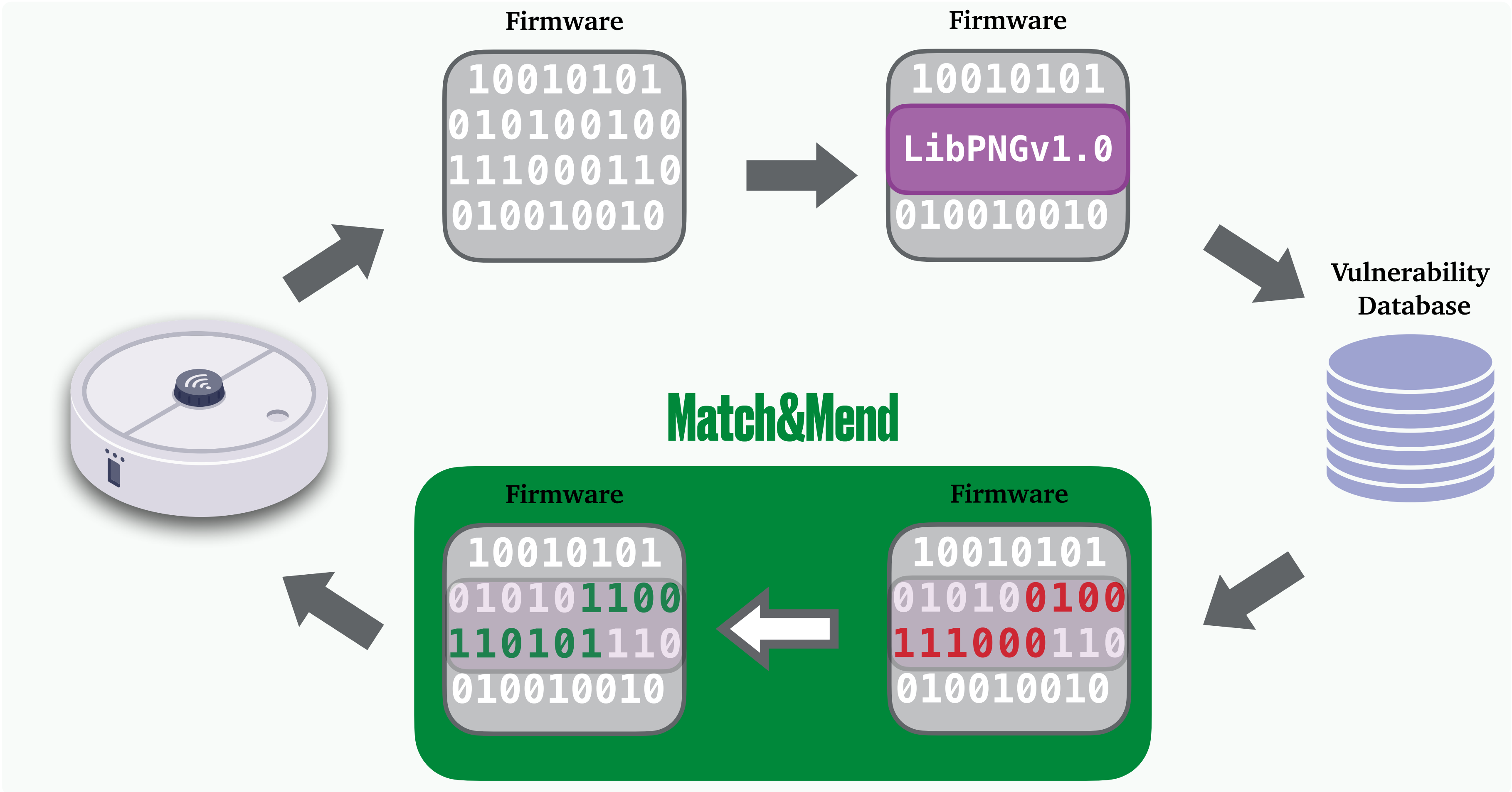


End-to-end patching pipeline



96%

FIRMWARE SUCCESS

126 of 131 patched libraries of real-world firmware

76%

MAGMA SUCCESS

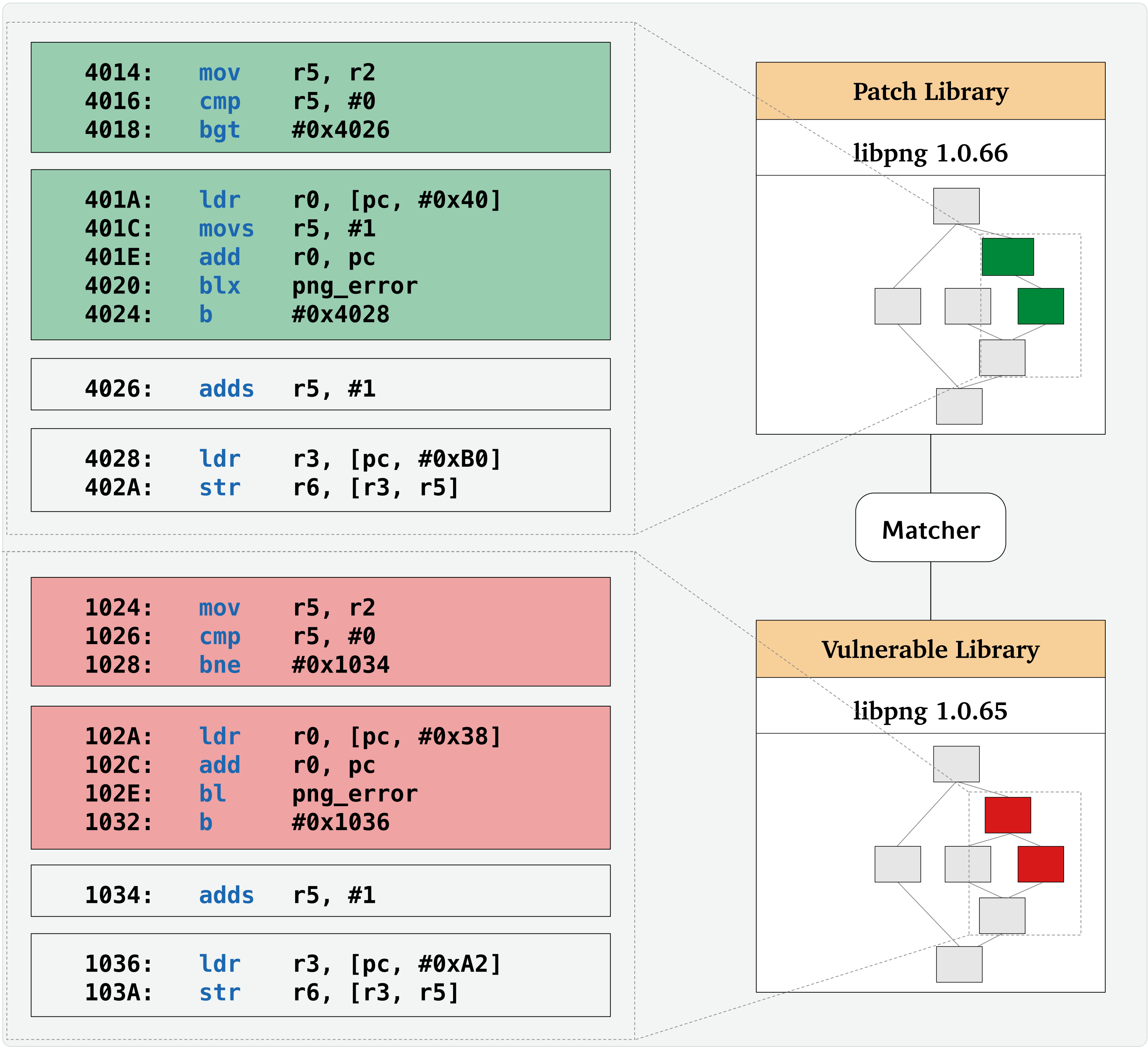
32 of 42 processed ARM32 benchmark libraries passed PoV/test validation.

5 – 46

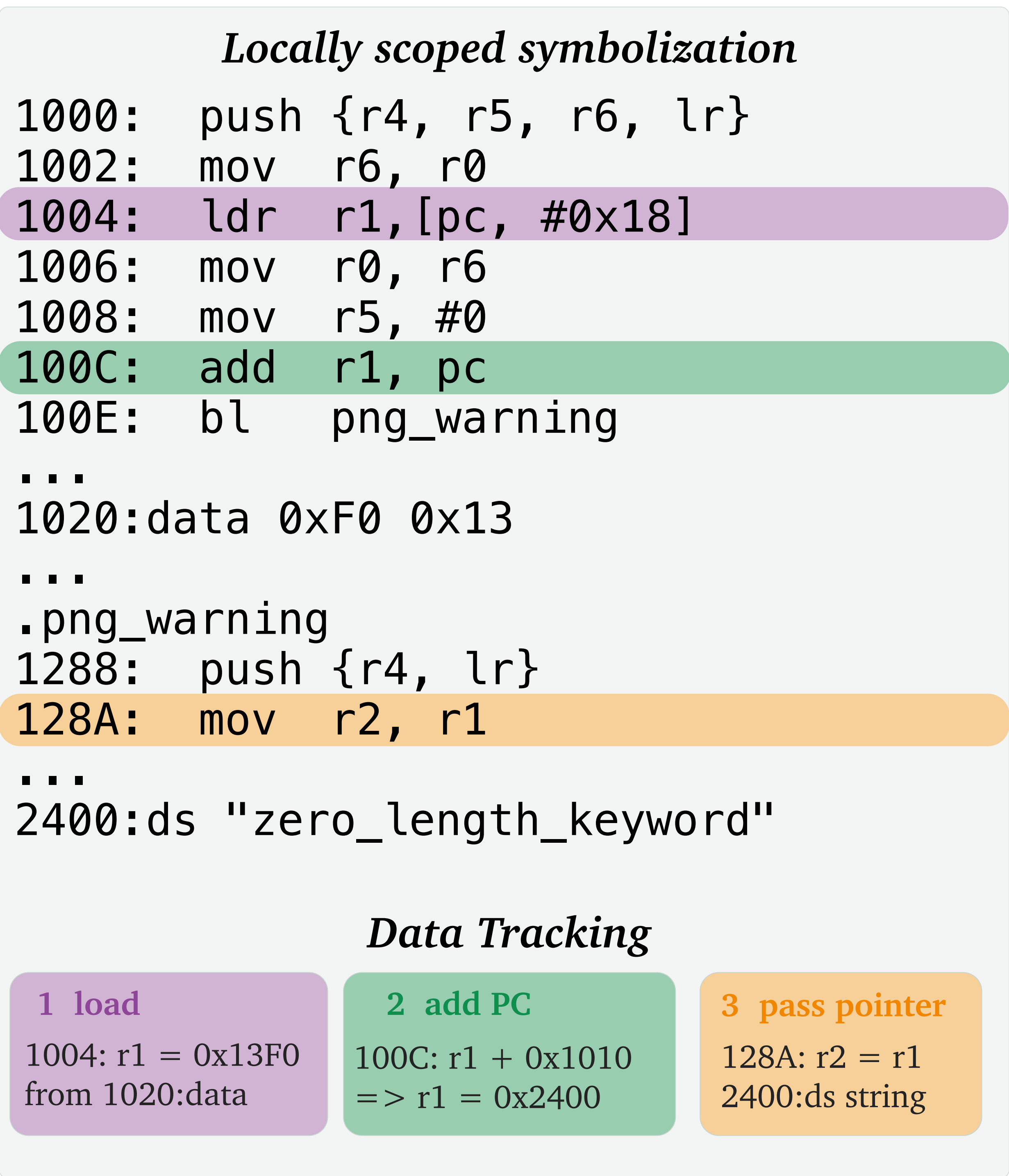
KIB PATCH SIZE

Observed average patch sizes stay small relative to affected libraries.

Match

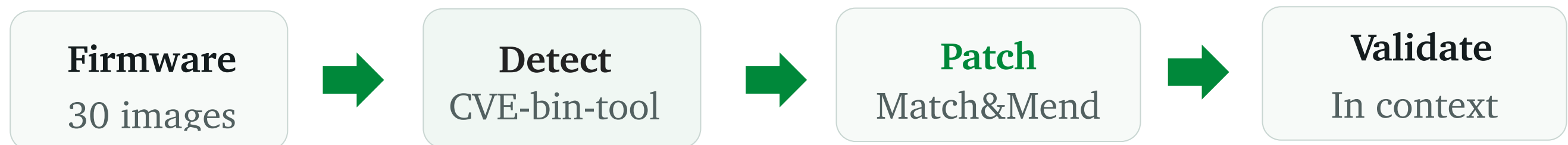


Mend



Real-world firmware (KARONTE dataset)

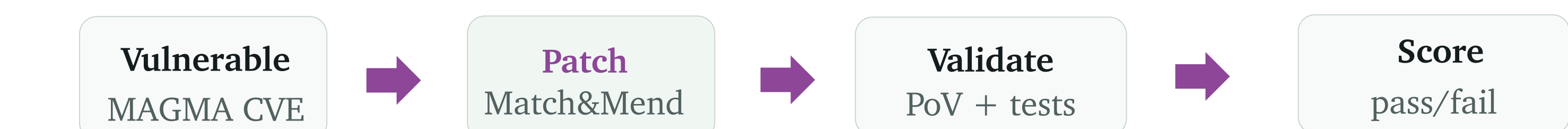
Checks whether the technique works inside real firmware images within their original filesystem context.



- Ground truth** CVE-bin-tool identifies vulnerable libraries in IoT firmware.
- Validation** Patched binaries run test suite in firmware context.
- Outcome** 126 of 131 patched libraries passed the tests.

Controlled MAGMA benchmark

Checks whether patching removes the known vulnerability while preserving normal library behaviour.



- Ground truth** Known vulnerable/patched code; ARM32 libraries built across O1/O2/O3.
- Validation** Proof-of-vulnerability stops triggering a crash; open-source test suite still passes.
- Outcome** 32 of 42 processed libraries passed, reported as 76% overall success.